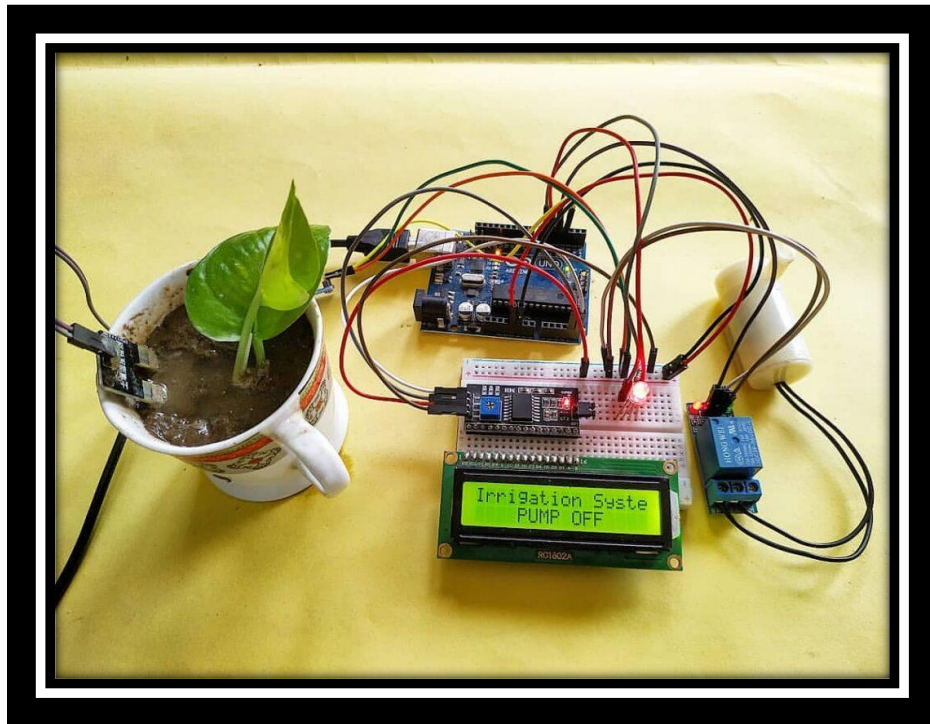


Automatic Plant Watering System



INTRODUCTION

A plant watering system using Internet of Things (IoT) technology revolutionizes the way we nurture and care for plants. Incorporating IoT into plant watering introduces a smart and automated approach to ensure optimal moisture levels, even when you're away. This system integrates sensors, connectivity, and automated mechanisms to monitor, analyze, and regulate watering, creating an efficient and effective way to maintain plant health. At its core, an IoT-enabled plant watering system typically comprises various components:

Sensors: These devices measure soil moisture, temperature, and sometimes even light exposure. They gather real-time data critical for understanding the plant's needs.

Connectivity: IoT systems use Wi-Fi, Bluetooth, or other wireless technologies to transmit data collected by sensors to a centralized control unit or a cloud-based platform.

Control Unit/Cloud Platform: This serves as the brain of the system, processing sensor data, executing predefined instructions or algorithms, and triggering actions such as watering based on the received information.

Actuators: These components execute the necessary actions based on the analyzed data. For instance, they could activate pumps to water the plants when moisture levels drop below a certain threshold.

Key benefits of an IoT-based plant watering system include:

Efficiency: Precise and timely watering based on real-time data, avoiding both overwatering and underwatering.

Remote Monitoring and Control: Access and manage the system from anywhere via a smartphone or web application, ensuring plant care even when you're away.

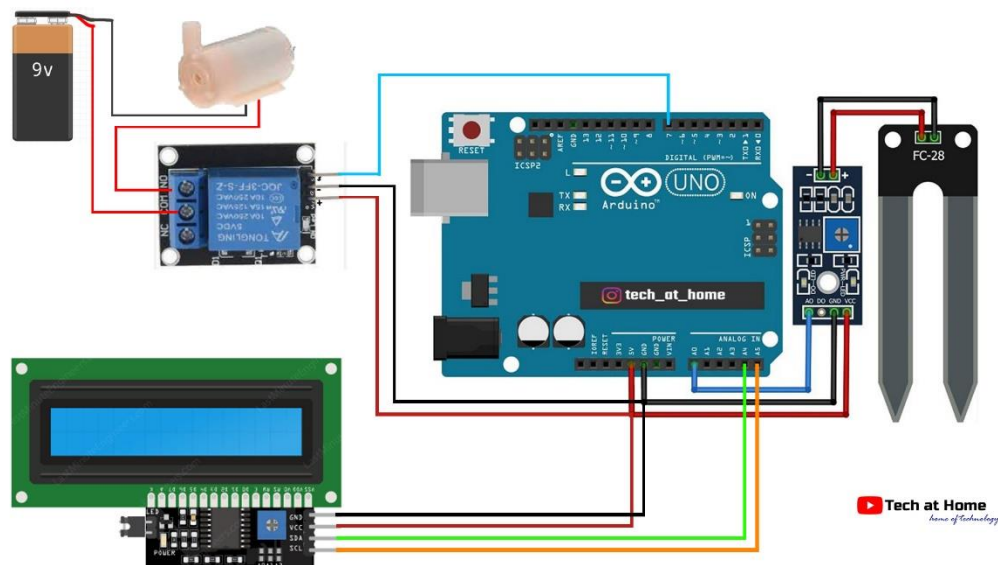
Conservation: Optimizing water usage by providing the necessary amount required by the plant, reducing wastage.

Plant Health Improvement: Creating an environment where plants thrive by maintaining ideal moisture levels.

Designing and implementing such a system involves understanding the specific needs of different plants, considering environmental factors, and fine-tuning the automated processes for optimal performance.

Additionally, ensuring security measures within the IoT infrastructure is crucial to protect data and prevent unauthorized access.

Overall, an IoT-based plant watering system represents a leap forward in plant care, offering a blend of technology and nature that makes nurturing plants more efficient, convenient.



APPARATUS

1. Soil Moisture sensor
2. Breadboard
3. Relay (9V)
4. Battery (9V)
5. Mini Motor Pump
6. LCD with i2c module
7. Arduino UNO
8. Jumper Wires

REAL LIFE APPLICATION OF THIS PROJECT-

Agriculture and Farming:

Precision Farming: Enable precise irrigation by analyzing soil conditions and weather data, conserving water and optimizing crop yield.

Large-Scale Management: Monitor and manage irrigation systems across expansive agricultural fields remotely, reducing manual labor.

Urban Landscaping and Green Spaces:

Efficient City Maintenance: Implement smart watering systems in parks, urban gardens, and green spaces to conserve water and maintain healthy landscapes.

Environmental Sustainability: Use IoT to reduce water waste and contribute to sustainable environmental practices in urban areas.

Eco-Friendly Initiatives:

Water Conservation: Contribute to global efforts in water conservation by using IoT-based systems to precisely deliver water to plants based on their needs.

Green Technology Adoption: Showcase the practical application of IoT in sustainable practices, encouraging adoption in other areas.

In essence, IoT plant watering systems find application in diverse settings, from personal gardens to large-scale agriculture, promoting efficient resource management, automation, and the cultivation of healthy plant life. These systems not only simplify plant care but also contribute to environmental sustainability and resource conservation.



CODE

```
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x27, 16, 2); //Initialise LCD

void setup() {
  Serial.begin(115200);
  lcd.init();
  lcd.backlight();
  lcd.clear();
  pinMode(2, OUTPUT); //Giving Pins
  digitalWrite(2, HIGH);
  delay(1000);
```



```
lcd.setCursor(0, 0);  
lcd.print("IoT Idyllic PUMP"); //For Intro  
lcd.setCursor(0, 1);  
lcd.print("Activated ");  
  lcd.print("");  
  delay(5000);  
lcd.clear();  
}
```

```
void loop() {  
  int value = analogRead(A0);  
  Serial.println(value);  
  if (value > 950) { //When soil is DRY  
    digitalWrite(2, LOW);  
    lcd.setCursor(0, 0);  
    lcd.print("Pump is ON");  
  } else {  
    digitalWrite(2, HIGH);  
    lcd.setCursor(0, 0);  
    lcd.print("Pump is OFF");  
  }  
}
```

```
}
```

```
if (value < 500) {
```

```
    lcd.setCursor(0, 1);
```

```
    lcd.print("Moisture : HIGH");
```

```
} else if (value > 500 && value < 950) { //When soil is semi-dry  
and semi-wet
```

```
    lcd.setCursor(0, 1);
```

```
    lcd.print("Moisture : MID ");
```

```
} else if (value > 950) { //When Soil is wet
```

```
    lcd.setCursor(0, 1);
```

```
    lcd.print("Moisture : LOW ");
```

```
}
```

```
}
```

CONCLUSION

A conclusion for an IoT-based plant watering system project could highlight its benefits, challenges faced, potential improvements, and the overall learning experience. Here's a sample conclusion:

In conclusion, the IoT-enabled plant watering system presents a practical solution for maintaining optimal soil moisture levels, ensuring the health and vitality of plants. Throughout this project, several key insights emerged.

Benefits:

Automated Maintenance: The system allows for automated watering, eliminating the need for constant manual monitoring.

Resource Efficiency: It optimizes water usage by supplying water only when the soil moisture drops below a predefined threshold, reducing waste.

Remote Monitoring: With IoT integration, remote monitoring and control become feasible, offering convenience and accessibility.

Challenges Faced:

Sensor Accuracy: Calibration and fine-tuning of the moisture sensor were crucial to ensure accurate readings and proper watering.

Component Compatibility: Ensuring compatibility among various components, such as pumps, sensors, and the IoT module, required careful consideration and testing.

Power Management: Efficient power management strategies were necessary to ensure the system's continuous and reliable operation.

Potential Improvements:

Enhanced Sensor Technology: Exploring more advanced sensors or incorporating multiple sensors could improve accuracy and reliability.

Mobile Application Integration: Developing a dedicated mobile app for real-time monitoring and control could enhance user experience and accessibility.

Data Analytics: Implementing data analytics to track plant growth patterns and optimize watering schedules based on historical data could further refine the system's efficiency.

Learning Experience:

This project provided invaluable hands-on experience in electronics, coding, IoT integration, and problem-solving. It deepened understanding in sensor calibration, hardware-software interaction, and the importance of iterative testing and refinement in IoT projects.

In essence, while the IoT-based plant watering system offers a functional solution to automated plant care, it remains a continuously evolving project with opportunities for enhancements and refinements to better cater to specific plant needs and user requirements.

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B.Tech 1st year IIoT Project
ECE(IoT) (A2)

